

 Marwadi University	Marwari University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing(01CT0513)	Aim: Simulate smoothing and sharpening operations on images using frequency domain filters.	
Open Ended Problem Experiment 9	Date: 04/09/2005	Enrollment No: 92301733054

Aim: Simulate smoothing and sharpening operations on images using frequency domain filters.

Programm:-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
def apply_gaussian_filter(image, sigma):
    image = np.float32(image)
    frequency_domain = cv2.dft(image, flags=cv2.DFT_COMPLEX_OUTPUT)
    shifted_frequency_domain = np.fft.fftshift(frequency_domain)
    rows, cols = image.shape
    crow, ccol = rows // 2, cols // 2
    mask = np.zeros((rows, cols, 2), np.float32)

    for i in range(rows):
        for j in range(cols):
            distance_squared = (i - crow) ** 2 + (j - ccol) ** 2
            gaussian_value = np.exp(-distance_squared / (2 * sigma**2))
            mask[i, j] = [gaussian_value, gaussian_value]

    filtered_frequency_domain = shifted_frequency_domain * mask
    shifted_filtered_frequency_domain = np.fft.ifftshift(filtered_frequency_domain)
    filtered_image = cv2.idft(
        shifted_filtered_frequency_domain, flags=cv2.DFT_SCALE | cv2.DFT_REAL_OUTPUT
    )
    filtered_image = cv2.normalize(filtered_image, None, 0, 255, cv2.NORM_MINMAX)
    return np.uint8(filtered_image)
def apply_sharpening_filter(image, strength):
    image = np.float32(image)
    frequency_domain = cv2.dft(image, flags=cv2.DFT_COMPLEX_OUTPUT)
    shifted_frequency_domain = np.fft.fftshift(frequency_domain)
    rows, cols = image.shape
    crow, ccol = rows // 2, cols // 2
    mask = np.ones((rows, cols, 2), np.float32)
    mask[crow - strength : crow + strength, ccol - strength : ccol + strength] = 0
    filtered_frequency_domain = shifted_frequency_domain * mask
    shifted_filtered_frequency_domain = np.fft.ifftshift(filtered_frequency_domain)
    filtered_image = cv2.idft(
        shifted_filtered_frequency_domain, flags=cv2.DFT_SCALE | cv2.DFT_REAL_OUTPUT
    )
```



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```
filtered_image = cv2.normalize(filtered_image, None, 0, 255, cv2.NORM_MINMAX)
return np.uint8(filtered_image)
image_paths = [
    "/content/smoothing data2.png",
    "/content/smoothing data3.png",
    "/content/smoothing data4.png",
    "/content/smoothing data5.png"
]
sigma = 20
strength = 20
for idx, path in enumerate(image_paths, start=1):
    input_image = cv2.imread(path, 0)
    if input_image is None:
        print(f"Could not load {path}")
        continue

    # Apply filters
    smoothed_image = apply_gaussian_filter(input_image, sigma)
    sharpened_image = apply_sharpening_filter(input_image, strength)

    # Display results
    plt.figure(figsize=(15, 5))
    titles = ["Original", "Smoothed (Gaussian)", "Sharpened (High-Pass)"]
    images = [input_image, smoothed_image, sharpened_image]

    for i in range(3):
        plt.subplot(1, 3, i + 1)
        plt.imshow(images[i], cmap="gray")
        plt.title(titles[i])
        plt.axis("off")

    plt.suptitle(f"Results for {path}", fontsize=14)
    plt.tight_layout()
    plt.show()

    # Save results
    cv2.imwrite(f"smoothed_image_{idx}.jpg", smoothed_image)
    cv2.imwrite(f"sharpened_image_{idx}.jpg", sharpened_image)

    print(f"Processed and saved results for {path}")
```



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Output:

Results for /content/smoothing data2.png

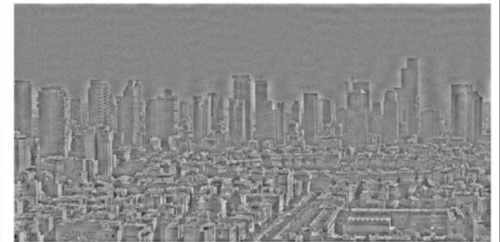
Original



Smoothed



Sharpened



Processed and saved results for /content/smoothing data2.png

Results for /content/smoothing data3.png

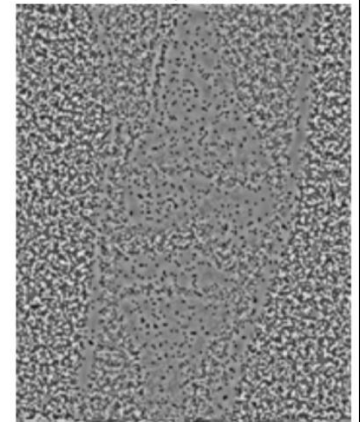
Original



Smoothed



Sharpened



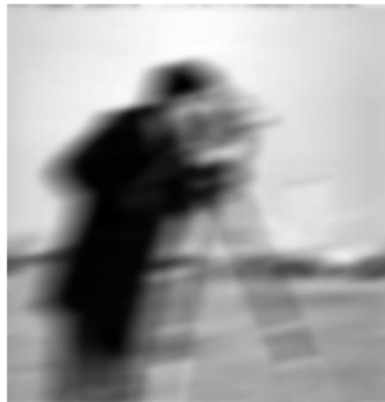
Processed and saved results for /content/smoothing data3.png

Results for /content/smoothing data4.png

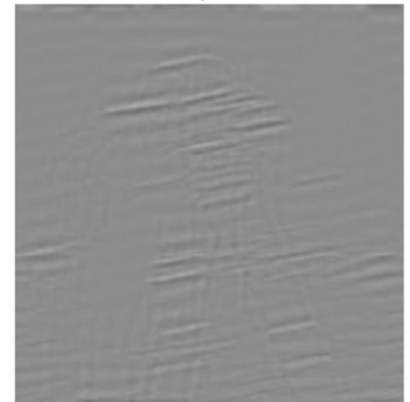
Original



Smoothed



Sharpened



Processed and saved results for /content/smoothing data4.png

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Observation:

- Gaussian smoothing ($\sigma = 20$) reduces high-frequency noise and smooths fine textures.
- Overall structure of images is preserved, but thin lines and small details are slightly blurred.
- Sharpening (strength = 20) enhances edges, boundaries, and prominent features.
- Noise in flat regions may be amplified and slight halos appear around strong edges.
- Across all four images, smoothing consistently cleans noisy areas.
- Sharpening consistently improves edge clarity and detail.
- The experiment highlights the trade-off between noise reduction and detail enhancement using frequency-domain filters.

Conclusion :-

From this experiment, we learn that smoothing effectively reduces noise while slightly blurring edges and fine details. Sharpening enhances edges and prominent features, making the image appear more defined, but it can also amplify noise and introduce minor halos. Overall, the experiment demonstrates that achieving optimal image quality requires carefully balancing smoothing and sharpening parameters based on the characteristics of each image.